

Explainable Fake News Detection using Retrieval-Augmented Generation

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Abstract—Fake news detection has become an important task due to the rapid spread of misinformation through online news and social media platforms. In this project, multiple Natural Language Processing (NLP) and machine learning techniques were explored to classify news articles as either fake or real based on their lexical style. The workflow begins with preprocessing and exploratory analysis of a public fake news dataset containing more than 44,000 articles collected from real-world sources. Two feature representation approaches, TF-IDF and Word2Vec embeddings, were then compared using Logistic Regression to determine the most effective feature extraction strategy. After selecting the best feature representation, three machine learning models, Logistic Regression, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP), were trained and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC metrics. Experimental results showed that Logistic Regression with TF-IDF features achieved the highest overall performance with an F1-score of 0.9937. To improve explainability, a Retrieval-Augmented Generation (RAG) pipeline was implemented using FAISS optimized similarity search and the FLAN-T5 language model to generate natural language explanations for model predictions. The final system combines high classification performance with interpretable prediction explanations.

Index Terms—Fake news detection, natural language processing, TF-IDF, Word2Vec, logistic regression, support vector machine, multi-layer perceptron, retrieval-augmented generation, FAISS, FLAN-T5, text classification, misinformation detection.

I. INTRODUCTION

The spread of fake news through online platforms has become a major challenge because misinformation influence public opinion, political discussions, and social behavior. News articles published online are also capable of quickly reaching millions of readers before their validity is verified by human intervention. Because of this, automated fake news detection systems have become increasingly important.

Machine learning and Natural Language Processing (NLP) techniques have shown strong performance in text classification tasks, making them suitable for detecting patterns commonly found in fake and real news articles. However, most of the fake news detection systems focus mainly on classification accuracy and provide little explanation for why an article was classified as fake or real. This can reduce user trust in the system.

In this project, several machine learning approaches were explored for fake news classification using NLP-based feature extraction techniques. The project compares TF-IDF and

Word2Vec feature representations and evaluates three machine learning models including Logistic Regression, Support Vector Machine (SVM), and Multi-Layer Perceptron (MLP). In addition, a Retrieval-Augmented Generation (RAG) pipeline was implemented to generate natural language explanations for predictions using retrieved similar articles from the training dataset and TF-IDF keyword analysis.

The main contributions of this project are:

- Performing exploratory analysis and dataset preprocessing.
- Comparing TF-IDF and Word2Vec feature extraction approaches.
- Evaluating multiple classification models using performance metrics.
- Implementing a Retrieval-Augmented Generation (RAG) explainability pipeline using FAISS and FLAN-T5.
- Generating human-readable explanations for predictions.

II. RELATED WORK

Fake news detection has been widely studied using Natural Language Processing and machine learning techniques. Wang, William Yang et al. [3] introduced a novel hybrid convolutional neural network (CNN) that integrates metadata with text, providing a new benchmark dataset to the field. This and many other works led to the application of traditional machine learning approaches to this problem. These approaches commonly use feature extraction methods such as TF-IDF [4] together with classifiers including Logistic Regression, Naive Bayes, and Support Vector Machines. TF-IDF is especially effective in text classification because it highlights the words and phrases that appear frequently in one document but rarely across the rest of the corpus, highlighting the crucial deciders of the text.

Word embeddings such as Word2Vec [5] have also been used for fake news detection. Unlike TF-IDF, Word2Vec captures semantic relationships between words by learning vector representations based on surrounding context. These dense vector representations are mainly used to capture meaning similarity between words appearing in different contexts.

More recently, explainability and interpretability have become important topics in machine learning systems. Retrieval-Augmented Generation (RAG) [6] systems combine retrieval techniques with language generation models with the goal of outputting grounded explanations based on retrieved evidence.

In this project, a lightweight RAG pipeline was implemented to provide evidence-based explanations for fake news predictions.

III. DATASET AND DATA PREPROCESSING

A. Dataset Description

The dataset used for this project was prepared by the ISOT Research lab at the University of Victoria. [1], [2] It contains two CSV files, Fake.csv and True.csv, representing fake and true articles respectively from 2016 to 2017. The true articles were gathered by web-crawling on reliable news sites like Reuters, whereas the fake news were gathered from articles flagged by Politifact.com. The datasets were loaded into separate dataframes and assigned labels where fake news samples were assigned label 0 and true news samples were assigned label 1. The two dataframes were then concatenated before analysis and preprocessing.

Dataset statistics are shown below:

- Fake News of 23,481 samples
- True News of 21,417 samples

The total samples in the dataset are 44,898.

The dataset contains the following columns:

- title - of the article.
- text - main body of the news.
- subject - category type.
- date - when the article was published.
- label - boolean value representing fake or true sample.

The final dataset shape after concatenation was (44898, 5) with a subject category distribution dominated by politics news, which is over 17,000 of the samples.

B. Initial Dataset Analysis

The data types of the features were explored before preprocessing. The title, text, subject, and date features were all stored as object types, while the label feature was numerical.

Additional features were created to analyze title and text lengths for each article. Basic statistics of the title and text lengths are shown below.

TABLE I: Descriptive Statistics for Text and Title Length

Statistic	Text Length	Title Length
Mean	2469.1	80.1
Standard Deviation	2171.6	25.4
Minimum	1.0	8.0
Median	2186.0	73.0
Maximum	51794.0	286.0

The dataset was also checked for missing values and duplicate samples. No null values were found in the dataset. However, 209 duplicate rows and 631 rows containing empty strings or whitespace-only content in either the title or text fields were identified. Since the dataset is large compared to these faulty rows, the rows were dropped.

C. Exploratory Data Analysis

Exploratory analysis plots were generated to study the distributions of the features and labels. The plots included:

- Class distribution between fake and real news samples.
- Title length distribution histograms.
- Text length distribution histograms.

Figure 1 shows a plot visualizing the dataset is relatively balanced between fake and real news classes. Figure 2 and Figure 3 display a class based character length frequency of the body, and title of articles, respectively, per class.



Fig. 1: Plot visualization of the dataset samples per class.

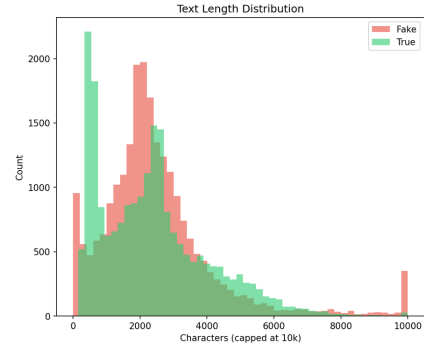


Fig. 2: Article text(body) length frequency display per class.

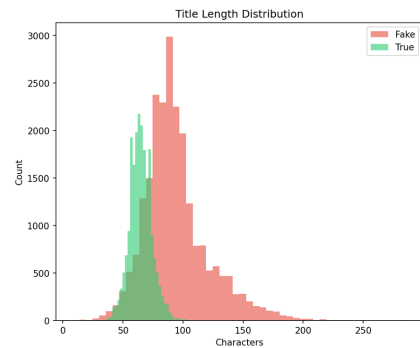


Fig. 3: Article title length frequency display per class.

From the above plot, it was observed that fake news samples generally had longer titles on average compared to real news samples. This suggested that title structure and writing style may provide useful information for classification.

D. Text Cleaning and Normalization

The purpose of cleaning the articles was to remove noise and reduce bias from irrelevant information before training the models. Several preprocessing techniques were applied:

- Standardizing text to lowercase.
- Removing hashtags, URLs, and user handles.
- Removing stopwords using NLTK library such as “is” and “and” that frequently appear but add little useful information.
- Lemmatization, also using NLTK, to reduce words to their root forms.
- Removing photo credits to reduce dataset-specific bias.

Lemmatization was useful for reducing words such as “running” and “ran” into their base word “run,” which improves word frequency analysis.

E. Training and Test Dataset Extraction

The dataset was divided into training and test sets using a stratified split. Stratification was used to preserve the original class distribution across both datasets.

The dataset was split into 75% training data and 25% test data.

- Training set of 17,136 Fake and 15,907 True labels, having a total of 33,043 samples.
- Test set of 5,712 Fake and 5,303 True labels, having a total of 11,015 samples.

Finally, the training and test datasets were shuffled and exported for carrying out the project’s experiment.

IV. METHODOLOGY

A. Feature Representation Techniques

Machine learning models require numerical feature representations before text data can be used for training. In this project, two feature extraction approaches were explored and compared:

- Term Frequency-Inverse Document Frequency (TF-IDF)
- Word2Vec embeddings

The purpose of this comparison was to evaluate the tradeoff between sparse statistical representation and dense semantic representation.

1) *TF-IDF Representation*: TF-IDF was used to capture words and phrases that appear frequently in one document but rarely across the full dataset. This approach generates a statistical representation of word importance.

A vocabulary of the 50,000 most common words and phrases was constructed and weighted using the TF-IDF formula. Both body-only text and combined title-plus-body text were evaluated.

L2 normalization was applied to the vectorization so that longer articles would not dominate simply because they contain more words.

The resulting feature matrices were:

- Body-only matrix: (33043, 50000)
- Title+Body matrix: (33043, 50000)

Even though TF-IDF extracts important meaning words, representing each sample using 50,000 vocabulary scores creates a sparse matrix where most values are zero because each article contains only a small subset of the vocabulary.

2) *Hyperparameter Tuning Using Cross Validation*: Cross validation was used to tune the hyperparameters for the inverse regularization strength (C) and the context window (ngram_range) while also comparing body-only input with title-plus-body input. This was done by training a Logistic Regression model and analyzing its performance for each configuration per 5 folds and taking the mean of the performance.

F1-score was selected as the primary performance metric because it provides information about both precision and recall. Mean and standard deviation values of F1-score were used to rank the hyperparameter configurations.

The best performing configurations are shown in the below table and in Figure 4 depicting the F1 results for each configuration.

TABLE II: Ranked model performance by varying regularization strength and N-Gram Range

Input	N-Gram Range	C	Mean F1	Std F1
Combined	(1,2)	10.0	0.992964	0.002445
Body	(1,2)	10.0	0.992308	0.002472
Combined	(1,2)	1.0	0.986987	0.002799
Body	(1,2)	1.0	0.986171	0.002810
Combined	(1,2)	0.1	0.970323	0.002755
Body	(1,2)	0.1	0.969985	0.003299
Body	(1,1)	0.1	0.967109	0.003370
Combined	(1,1)	0.1	0.967064	0.003019

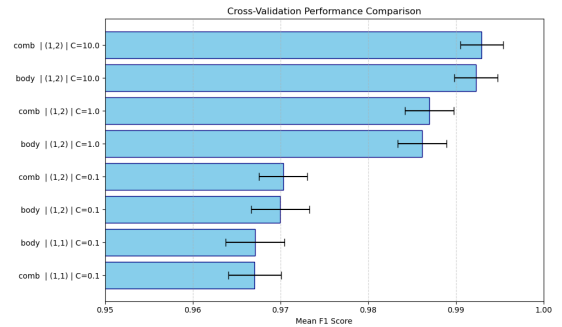


Fig. 4: Hyperparameter tuning cross validation performance.

The above plot shows the combined title and body representation slightly outperformed body-only feature.

3) *Word2Vec Embeddings*: Word2Vec embeddings were explored as an alternative feature representation approach. In this method, words are represented using 100-dimensional vectors learned from neighboring word contexts. A context window size of 5 words was used during training.

The Word2Vec model was also trained only on the training dataset, avoiding data leakage.

A document vector construction function was implemented to combine individual word vectors into a single 100-dimensional representation for each article.

The resulting matrices were:

- W2V body matrix: (33043, 100)
- W2V combined matrix: (33043, 100)

B. Feature Representation Comparison

The best TF-IDF configuration was compared against Word2Vec embeddings using Logistic Regression on unseen test data.

TABLE III: Model Performance by Representation and Input

Representation	Input	Accuracy	F1
TF-IDF	Body-only	0.9953	0.9951
TF-IDF	Title+Body	0.9939	0.9937
Word2Vec	Title+Body	0.9757	0.9749
Word2Vec	Body-only	0.9680	0.9670

The TF-IDF representation achieved the highest overall performance even though it produces sparse vectors. This is likely because TF-IDF effectively captures important words and phrases that strongly distinguish fake and real news articles.

In the unseen data comparison, the performance difference between body-only and combined title-plus-body input was relatively small, but body-only had a higher performance. However, since the combined representation preserves additional contextual information, the title and body features were kept together for later experiments.

Finally, the selected TF-IDF vectorizer and best configuration were exported for use during model training.

C. Model Training

1) *Feature Matrix Construction*: The selected TF-IDF vectorizer was used to transform the training and test datasets into sparse feature matrices.

- X_train shape: (33043, 50000)
- X_test shape: (11015, 50000)

The TF-IDF vectorizer scanned every article in the training dataset and built a fixed vocabulary containing the 50,000 most common words and phrases. Each article was then represented using the TF-IDF scores of these vocabulary terms.

Since most articles contain only a small subset of the vocabulary, most feature values for each sample are zero, resulting in a sparse matrix representation.

An example of the top weighted TF-IDF features for the first training sample is shown below.

2) *Logistic Regression*: Logistic Regression was first evaluated because it commonly performs well on high-dimensional sparse text data. The model was trained using the selected TF-IDF feature representation and is configured with the inverse regularization strength, $C=10$, using the lbfgs solver with a maximum of 1000 iterations to ensure convergence. Figure 5 represents the confusion matrix and standard classification metrics for the model.

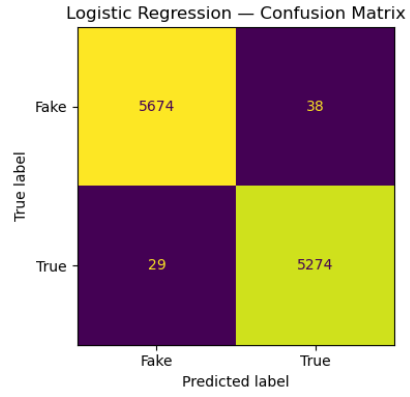


Fig. 5: Confusion Matrix for the Logistic Regression model.

3) *Support Vector Machine (SVM)*: An SVM model with an RBF kernel was also evaluated. Due to the large dataset size and the high dimensionality of the 50,000-column TF-IDF matrix, TruncatedSVD was used to reduce the sparse feature representation to 300 dimensions before training. Figure 6 represents the confusion matrix for the SVM.

This dimensionality reduction step significantly reduced computational cost while preserving the most important information from the feature space.

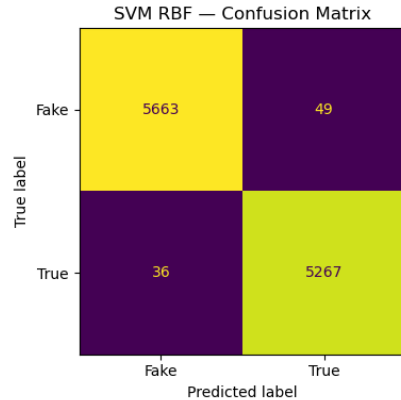


Fig. 6: Confusion Matrix for the SVM model.

4) *Multi-Layer Perceptron (MLP)*: The MLP consists of a three-layer feed-forward architecture (256 and 128 hidden units) employing ReLU activation functions and a 0.3 dropout rate to prevent overfitting. It is trained using the Adam optimizer with a learning rate of 10^{-3} for 20 epochs and Binary Cross-Entropy (BCE) as the objective loss function. Figure 7 represents the confusion matrix and standard classification metrics for the MLP.

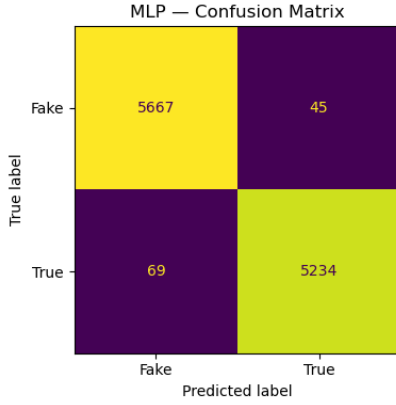


Fig. 7: Confusion Matrix for the MLP model.

V. EXPERIMENTAL RESULTS

A. Model Comparison

The performance of the three evaluated models is shown below. Figure 8 represents a bar chart for each metric per model, visualizing the differences and Figure 9 represents the ROC curves of each model in a single plot, showing the performance difference.

TABLE IV: Performance Comparison of Machine Learning Models

Model	Accuracy	Precision	Recall	F1	ROC-AUC
LR	0.9939	0.9928	0.9945	0.9937	0.9996
SVM (RBF)	0.9923	0.9908	0.9932	0.9920	0.9992
MLP	0.9897	0.9915	0.9870	0.9892	0.9989

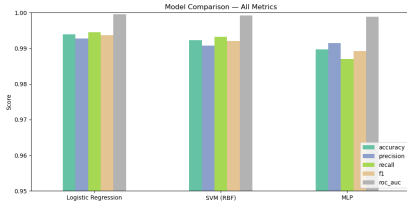


Fig. 8: Model performance metrics comparison.

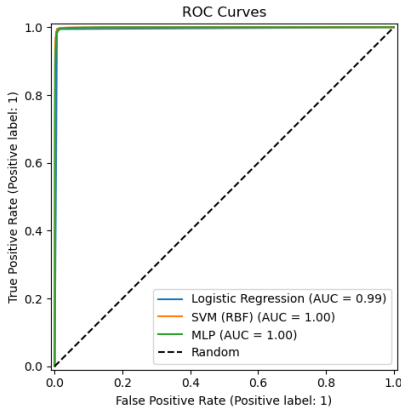


Fig. 9: ROC-AUC plot for the three models.

The models achieved very similar performance overall. However, Logistic Regression achieved the highest performance across all evaluation metrics.

The strong performance of Logistic Regression suggests that the TF-IDF feature space is highly effective for separating fake and real news articles using a linear classifier. Sparse TF-IDF representations often work well with linear models because important words and phrases become highly discriminative features.

The SVM model also performed strongly, but training required dimensionality reduction because of the computational cost associated with the large sparse feature matrix. The MLP neural network achieved slightly lower performance compared to the other approaches

B. Discussion of Results

The overall results indicate that TF-IDF feature extraction combined with traditional machine learning models performs extremely well for this fake news classification task.

One possible reason for the very high performance metrics is that fake and real news articles in the dataset may contain strong stylistic differences, recurring phrases, topic patterns, and title length differences that are highly distinguishable using statistical word-frequency methods.

Although Word2Vec embeddings capture semantic relationships between words, the TF-IDF approach produced stronger performance because important discriminative words and phrases appear to play a significant role in distinguishing fake and real news articles.

The final selected model was Logistic Regression using TF-IDF features.

VI. EXPLAINABILITY USING RAG

A. Motivation for Explainability

While classification accuracy is important, explainability is also necessary in fake news detection systems. Users may not trust predictions if the system only outputs a label without supporting evidence or reasoning.

To improve interpretability, a Retrieval-Augmented Generation (RAG) pipeline was implemented to generate natural language explanations for predictions by retrieving similar training articles and important TF-IDF phrases.

B. FAISS Retrieval Pipeline

The training dataset was vectorized using the selected TF-IDF vectorizer and indexed using Facebook AI Similarity Search (FAISS) [7].

The main purpose of the retrieval system was to efficiently search for articles in the training dataset that are similar to a given test article. Performing brute-force cosine similarity search across the full dataset for every prediction would be computationally expensive.

FAISS was used to optimize nearest-neighbor retrieval in the high-dimensional TF-IDF feature space.

The constructed FAISS index contained indexed vectors of 33,043, each with a dimension of 50,000.

The retrieval pipeline performs the following steps:

- 1) Vectorize the input news article using the TF-IDF vectorizer.
- 2) Retrieve the top-k (taken value of $k=3$) most similar training samples using FAISS.
- 3) Extract high-scoring TF-IDF phrases from the input article.
- 4) Construct a natural language prompt.
- 5) Generate an explanation using FLAN-T5.

Figure 10 below represents the actions performed for each step of the RAG pipeline.

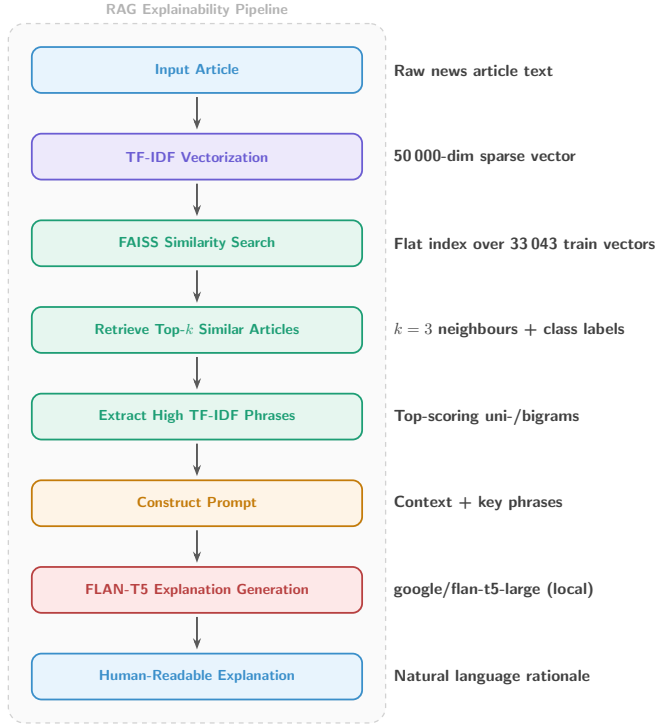


Fig. 10: RAG Pipeline flowchart.

C. Key Phrase Extraction

A TF-IDF phrase extraction function was implemented to identify important words and phrases with high TF-IDF scores from the input article.

These extracted phrases were used during similarity analysis and explanation generation because they represent the features most strongly influencing the classifier.

D. Natural Language Explanation Generation

To generate human-readable explanations, the pretrained language model google/flan-t5-large was downloaded from HuggingFace and executed locally.

The classifier itself does not directly generate explanations. Instead, the RAG system performs the explanation generation after the prediction by combining:

- Retrieved similar articles
- Neighboring article labels

- High-scoring TF-IDF phrases
- Prompt-based language generation

This approach grounds the generated explanations using retrieved evidence from the training dataset.

E. Example Explanation Output

An example prediction and generated explanation is shown below.

True label: REAL

Prediction: REAL

Generated explanation:

"This article is classified as REAL with 99.9% confidence. The most distinctive phrases found were: 'muslimmajority country', 'muslimmajority', 'trump', 'refugee people', 'order'. These phrases appear frequently in REAL articles in the training data. 3 out of 3 most similar training articles were also labeled REAL, further supporting this prediction."

The generated explanation combines retrieved evidence and important phrases to provide a human-readable justification for the prediction.

One limitation of this approach is that explanation quality depends heavily on retrieval quality. The assumption made here is that relevant articles will always be retrieved. If unrelated articles are retrieved, the generated explanation may become less reliable.

VII. CONCLUSION AND FUTURE WORK

In this project, multiple NLP and machine learning techniques were explored for fake news detection. The workflow included dataset preprocessing, exploratory data analysis, feature extraction, model comparison, and explainability generation.

TF-IDF and Word2Vec feature representations were compared, and TF-IDF achieved the strongest overall performance. Three machine learning models, Logistic Regression, SVM, and MLP, were evaluated using several classification metrics. Logistic Regression combined with TF-IDF features achieved the highest performance with an F1-score of 0.9937.

To improve interpretability, a Retrieval-Augmented Generation (RAG) pipeline was implemented using FAISS similarity retrieval and FLAN-T5 natural language generation. The explainability system generated evidence-based natural language explanations using retrieved similar articles and important TF-IDF phrases.

Future improvements could include testing transformer-based classifiers such as BERT [8], expanding the explainability pipeline with more advanced retrieval methods, and evaluating the system on more diverse and real-time news datasets.

APPENDIX

GitHub Repository reference:- <https://github.com/natitedros/explainable-fake-news-detection/>

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